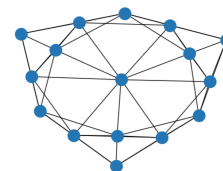


Kevin Schmidt

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🔗 kevinschmidt24799



I am a math graduate student interested in spectral graph theory, linear algebra, combinatorics and representation theory.

Education

- 2024–2026 **M.A. in Mathematics**, *San Jose State University*, GPA 3.9
Thesis title: Upper Bounds on Third Eigenvalues of Graphs
- 2023–2024 **B.A. in Mathematics**, *San Jose State University*, GPA 3.9
- 2021–2023 **A.A. in Mathematics**, *Foothill Community College*, GPA 3.6

Awards

- 2025 Franklin B. Fuller Scholarship
- 2025 Hearthstone Summer Championship Qualifier
- 2024–2025 SJSU President's Scholar
- 2023–2024 SJSU President's Scholar
- 2024 SJSU Magna Cum Laude, Latin Honor & Medallion
- 2023 SJSU Richard C. Dieckmann Scholarship for Academic Excellence

Research Experience

- 8/2025–present **Master's Thesis with Professor Wasin So in spectral graph theory, SJSU**
- Working on bounding the third-largest eigenvalue of graphs and characterizing extremal examples. A result of Powers claims that $\lambda_3(G) < |G|/3$, but this had an error and remains open.
 - Found families of graphs which cannot serve as counterexamples.
 - Proved eigenvalue bounds and structural properties that a counterexample graph must have, including restrictions on their automorphism groups.
- 1/2024–5/2024 **Research with Professor Bering in geometric group theory, SJSU**
- Successfully proved no finite cover of the fundamental group of the figure 8 knot complement $\pi_1(4_1)$ has a Dehn complex that is special. This was previously only known for covers of index greater than 10.
 - Computationally evaluated cube complexes for osculation.
 - Wrote an exhaustive search in Python using the low index module.

Work Experience

- 8/2024–12/2024 **Graduate Teaching Associate for Discrete Mathematics, SJSU**
- I worked as a graduate teaching associate, designing and running workshops (support classes) for undergraduate discrete math students.
 - This included compiling suitable practice problems that reinforced core concepts while developing students' proof and exposition skills.
 - I received a 5/5 rating from my students; feedback included "He was very helpful and went deep into the concept he was explaining", "he would explain it and make sure I understood", and "This instructor was frequently accessible on Discord".
- 8/2020–8/2024 **Private Mathematics Tutor**
- Independently, I worked as a private tutor for middle and high school students, teaching geometry, trigonometry, calculus, and competition math. I had both advanced and remedial students, so I had to diagnose students' specific capabilities and gaps to adjust my approach accordingly.

Other skills

Fluent in both English and Mandarin
Skilled with Python, C/C++, LaTeX, Git

Research-related software projects

Fast Graph Eigenvalue Search

- This was a highly-optimized C++ search of all 165 billion 12-vertex graphs unique up to isomorphism. It took 33 hours to run on a Macbook Pro M2.
- <https://github.com/kevinschmidt24799/12-vertices-large-third-eig/blob/main/RESULTS.md>

Blowup Search

- This Python program searched graphs with a threefold or fivefold symmetry. More precisely, graphs G such that $\text{Aut}(G)$ contained an element of order 3 or 5 respectively. For such graphs, it evaluated if the closed blowup of the graph $G^{[t]}$ could have large third eigenvalue.
- <https://github.com/kevinschmidt24799/blowup-sym>

Covers of Figure-8 Knot Complement

- With Professor Edgar Bering, I wrote a Python program which searched small covers of the fundamental group of the figure-8 knot complement. For covers of index at most 10, we checked the Dehn cell complex for different types of osculation and characterized whether the cover was special.
- <https://github.com/ebering/figure-8-covers>

Math coursework

Spring 2026	Higher Algebra II (Math 221B), <i>Edgar Bering</i>	In progress
Spring 2026	Fourier Analysis (280), <i>Kyle Hambrook</i>	In progress
Spring 2026	Master's Thesis (Math 299), <i>Wasin So</i>	In progress
Fall 2025	Higher Algebra I (Math 221A), <i>Edgar Bering</i>	A
Fall 2025	Topology (Math 275A), <i>Yan Zhang</i>	A
Fall 2025	Master's Thesis (Math 299), <i>Wasin So</i>	P
Spring 2025	Functional Analysis (Math 231B), <i>Slobodan Simic</i>	A+
Spring 2025	Theory of Numbers (Math 226), <i>Jordan Schettler</i>	A+
Spring 2025	Guided Graduate Individual Studies (Math 280), <i>Wasin So</i>	P
Spring 2025	Advanced Topics in Mathematics (Math 285M), <i>Wasin So</i>	A
Fall 2024	Graph Theory (Math 279A), <i>Yan Zhang</i>	A+
Fall 2024	Real Analysis (Math 231A), <i>Slobodan Simic</i>	A
Fall 2024	Advanced Matrix Theory (Math 229), <i>Wasin So</i>	A-
Spring 2024	Theory of Numbers (Math 126), <i>Jordan Schettler</i>	A+
Spring 2024	Abstract Algebra II (Math 128B), <i>Edgar Bering</i>	A
Spring 2024	Introduction to Analysis (Math 131A), <i>Timothy Hsu</i>	A
Spring 2024	Linear Algebra II (Math 129B), <i>Wasin So</i>	A-
Spring 2024	Undergraduate Research (Math 180R), <i>Edgar Bering</i>	P
Fall 2023	Introduction to Graph Theory (Math 179), <i>Wasin So</i>	A
Fall 2023	Applied Probability and Statistics (Math 161A), <i>Cristina Tortora</i>	A
Fall 2023	Introduction to Combinatorics (Math 142), <i>Wasin So</i>	A
Fall 2023	Abstract Algebra I (Math 128A), <i>Edgar Bering</i>	A
Spring 2023	Intro to Abstract Math and Proofs (Math 108), <i>Kyle Hambrook</i>	A-
Fall 2022	Complex Variables (Math 138), <i>Giang Le</i>	A

Math coursework (with textbook and descriptions)

- Spring 2026 **Higher Algebra II (Math 221B)**, *Edgar Bering* In progress
- Text: Algebra: Chapter 0, Paolo Aluffi
 - Description: Hilbert spaces, Banach algebras, operator theory, spectral theory of operators.
- Spring 2026 **Fourier Analysis (280)**, *Kyle Hambrook* In progress
- Text: An Introduction to Harmonic Analysis, Katznelson. Fourier Series and Integrals, Dym & McKean. Fourier Analysis: An Introduction, Stein & Shakarchi.
 - Description: Fourier series and transform on the circle and line, Weyl equidistribution, Hardy Tauberian theorem, Riemann localization, Schwartz class, Riemann-Lebesgue inversion, applications to ODEs, heat equation, discrete transform on groups.
- Spring 2026 **Master's Thesis (Math 299)**, *Wasin So* In progress
- Continuation of Master's thesis.
- Fall 2025 **Higher Algebra I (Math 221A)**, *Edgar Bering* A
- Text: Algebra: Chapter 0, Paolo Aluffi
 - Description: Category theoretical approach to groups, rings, integral domains, modules, fields, vector spaces. Products, coproducts, kernels, cokernels, quotients as universal properties, exact sequences, split sequences, the snake lemma.
- Fall 2025 **Topology (Math 275A)**, *Yan Zhang* A
- Text: Topology without Tears, Sidney A. Morris
 - Description: Topological spaces and associated concepts (e.g., subspaces, product spaces, quotient spaces); continuous functions; compactness, connectedness (including path connectedness) and their local versions; countability and separation axioms; compactifications and Tychonoff's Theorem; paracompactness and metrization theorems.
- Fall 2025 **Master's Thesis (Math 299)**, *Wasin So* P
- Description: Spectral graph theory, working on bounding the third largest eigenvalue of graphs and characterizing extremal examples. A result of Powers claims that this eigenvalue is at most a third of the graph order, but this had an error. Found new families of graphs which cannot serve as counterexamples, and eigenvalue bounds that a theoretical counterexample graph must have.
- Spring 2025 **Functional Analysis (Math 231B)**, *Slobodan Simic* A+
- Text: Introduction To Hilbert Spaces with Applications, Lokenath Debnath
 - Description: Function spaces and their duals, operators on function spaces, Hilbert spaces, Banach algebras, operator theory, spectral theory of operators
- Spring 2025 **Theory of Numbers (Math 226)**, *Jordan Schettler* A+
- Text: Introduction to Analytic Number Theory, Tom M. Apostol
 - Description: Prime number theorem, Mobius inversion, Riemann zeta function, Dirichlet series, L function, Circle method, projective plane, partitions.
- Spring 2025 **Guided Graduate Individual Studies (Math 280)**, *Wasin So* P
- Text: The Probabilistic Method, Spencer & Alon. Additive Combinatorics, Tao & Vu
 - Description: Individual study in a specific field. The probabilistic method, extremal graph theory, Ramsey theory, additive combinatorics, discrete isoperimetric problems.
- Spring 2025 **Advanced Topics in Mathematics (Math 285M)**, *Wasin So* A
- Topic: Nonnegative Matrix Theory and Applications
 - Text: Matrix Analysis, Horn and Johnson
 - Description: Perron Frobenius theory of positive, primitive, irreducible, and non-negative matrices, M matrices, eigenvalue bounds, spectral graph theory, spectral moments.
- Fall 2024 **Graph Theory (Math 279A)**, *Yan Zhang* A+
- Text: Introduction to Graph Theory, Douglas B. West
 - Description: Graphs, digraphs, trees, graph embeddings, matchings, spanning trees, topological sorts, factorizations, colorings, Ramsey theory, spectral graph theory, adjacency and Laplacian matrices

- Fall 2024 **Real Analysis (Math 231A)**, *Slobodan Simic* A
- Text: Real Analysis: Measure Theory, Integration, and Hilbert Spaces (Princeton Lectures in Analysis), Stein and Shakarchi
 - Description: Sigma algebras, construction of measures, Lebesgue measure, measurable functions, differentiation, integration theory, convergence theorems, Riesz representation theorem, probability theory.
- Fall 2024 **Advanced Matrix Theory (Math 229)**, *Wasin So* A-
- Text: Matrix Analysis, Horn and Johnson
 - Description: Eigenvalues, unitary equivalence and Schur's theorem. Normal, Hermitian and symmetric real matrices. Positive definite matrices, polar and singular value factorizations, and selected topics at the discretion of the instructor.
- Spring 2024 **Theory of Numbers (Math 126)**, *Jordan Schettler* A+
- Text: Elementary Number Theory, David M. Burton
 - Description: Divisibility, prime numbers, congruences of first and higher degrees, theorems of Fermat, Euler and Wilson. Quadratic residues.
- Spring 2024 **Abstract Algebra II (Math 128B)**, *Edgar Bering* A
- Text: Discovering Abstract Algebra, John K. Osoinach Jr.
 - Description: Emphasis on rings, integral domains, fields, field extensions, Galois theory.
- Spring 2024 **Introduction to Analysis (Math 131A)**, *Timothy Hsu* A
- Text: Elementary Analysis: The Theory of Calculus, Kenneth A. Ross
 - Description: Properties of real numbers including completeness and compactness. Continuous functions, uniform continuity, the derivative.
- Spring 2024 **Linear Algebra II (Math 129B)**, *Wasin So* A-
- Text: Linear Algebra, Friedberg, Insel, and Spence
 - Description: Continuation of MATH 39. Abstract vector spaces and linear transformations, diagonalization, Cayley-Hamilton theorem, minimal polynomials, Jordan canonical form. Selected topics from inner product and adjoint, duality, rational canonical form and applications.
- Spring 2024 **Undergraduate Research (Math 180R)**, *Edgar Bering* P
- Text: The Symmetries of Things, John H. Conway. Office Hours with a Geometric Group Theorist, Clay and Margalit
 - Description: Met weekly with the professor. Explored group theory, fundamental groups, knots, Dehn complexes, flag complexes, special covers. Wrote a Python program to successfully prove a novel result via computation.
- Fall 2023 **Introduction to Graph Theory (Math 179)**, *Wasin So* A
- Text: Introduction to Graph Theory, Robin Wilson
 - Description: Hamiltonian and Eulerian properties, matching, trees, connectivity, coloring problems and planarity. Emphasis on algorithms and applications, including optimal network flows.
- Fall 2023 **Applied Probability and Statistics (Math 161A)**, *Cristina Tortora* A
- Text: Professor's slides and notes
 - Description: Descriptive and inferential statistics. Collection and analysis of data, discrete and continuous probability models, random variables, Central Limit Theorem, confidence intervals, hypothesis testing.
- Fall 2023 **Introduction to Combinatorics (Math 142)**, *Wasin So* A
- Text: Applied Combinatorics, Alan Tucker
 - Description: Sets, permutations, combinations, probability, mathematical induction, counting techniques, generating functions, partitions, recurrence relations, inclusion-exclusion. Polya's theorem and applications to computer science, mathematics, engineering and physical sciences.
- Fall 2023 **Abstract Algebra I (Math 128A)**, *Edgar Bering* A
- Text: Discovering Abstract Algebra, John K. Osoinach Jr.
 - Description: Group theory: permutation groups, abelian groups, morphism theorems, finite groups. Introduction to rings and fields.

- Spring 2023 **Intro to Abstract Math and Proofs (Math 108)**, *Kyle Hambrook* A-
- Text: Assorted materials assembled by professor
 - Description: Develop students' mathematical maturity and skill with proofs.
 - Material includes logic; set theory including functions, relations, and cardinality; the real number system, including the completeness axiom; and selected topics.
- Fall 2022 **Complex Variables (Math 138)**, *Giang Le* A
- Text: Fundamentals of Complex Analysis, Saff and Snider
 - Description: Analytic functions, complex integration, residues and power series.